## Fully coupled concentration dynamics

$$\underbrace{\frac{\partial \phi}{\partial t}}_{\text{Concentration variation}} = \underbrace{-\nabla \cdot (\boldsymbol{u}\phi)}_{\text{Advection}} + \underbrace{\nabla \cdot \left(\Gamma_\phi \nabla \frac{\delta \mathcal{F}}{\delta \phi}\right)}_{\text{Cell sorting}} + \underbrace{\nabla \cdot \left(\sqrt{2\Lambda}\zeta\right)}_{\text{Cell motion noise}} + \underbrace{S_\phi}_{\text{Cell differentiation}}$$

Diffusive time scale $\tau_D = \dfrac{L^2\alpha}{\Gamma_\phi \sigma_{12}}$

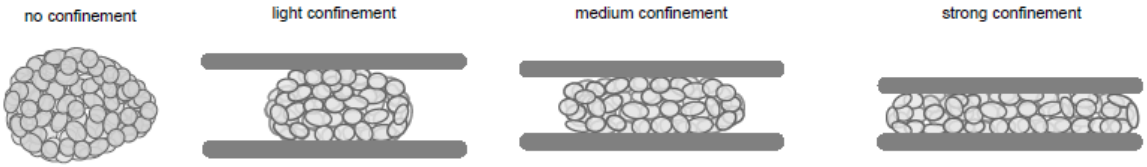
Noise time scale $\tau_N = \dfrac{L}{\sqrt{\Lambda}}$

Inverse of relative noise amplitude

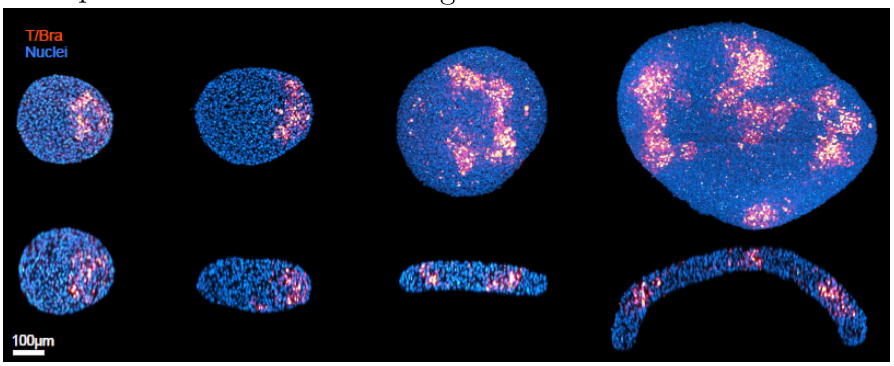
$$\frac{\tau_N}{\tau_D} = \frac{\sigma_{12}}{\alpha L}\frac{\Gamma_\phi}{\sqrt{\Lambda}}$$

Source term

$$S_\phi = A(1-\psi)(1-\phi)$$



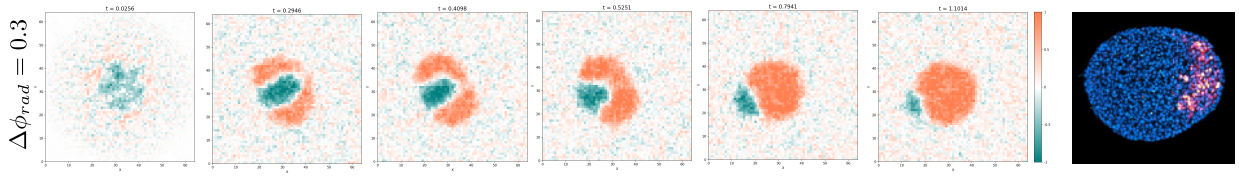

Dipole formation Ring formation Cluster formation



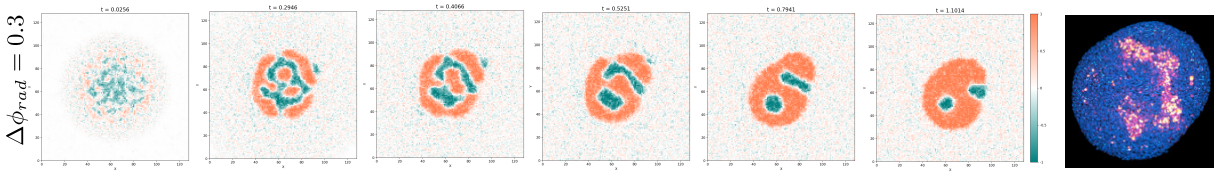

$Pe = \tau_D/\tau_{adv}$

$Da = A\tau_D$

$L = 32$ $Da = 3.125$ $Pe = 25$ $\Delta\phi_{rad}$=Strength of initial radial gradient




$L = 64$ $Da = 12.5$ $Pe = 50$




$L = 128$ $Da = 50$ $Pe = 100$

